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	Industry Project InnaMed is leading the integration of molecular information into home healthcare by innovatively combining microneedle blood collection, electrochemical biomarker quantification technology, and cloud-based data transfer in a single, portable device called TeleLab. Serial and frequent blood testing is a foundation of outpatient care to assess stability or change in key blood analytes to guide therapeutic decision-making and to evaluate organ dysfunction. However, only a small portion of blood testing may be performed at home, such as painful finger stick glucose testing, with the bulk being performed at outpatient laboratories, thereby requiring patient willingness, effort, time, and cost for travel and waiting, as well as additional caregiver support, in order to comply with recommended follow-up clinic visits. The vision for TeleLab is to deliver clinical value by diminishing blood testing as a reason for clinic visits, thereby		

Task Description:	shifting more outpatient care to the home setting as a component of frequent and convenient e-visits with practitioners via electronic media. Each function of TeleLab addresses critical opportunities to improve outpatient care. The rapid and virtually painless sampling of a few drops of blood from the skin capillaries by a microneedle patch removes anxiety and discomfort as obstacles to patient self-testing, thereby improving compliance. Our published, patented and validated electrochemical proximity assay and DNA nanostructure technologies enable rapid (<10 min), ultrasensitive (fM LoD) measurement of a broad variety of blood analytes, including electrolytes (such as sodium or potassium), drug molecules, peptides and proteins, using novel applications of electrochemistry and aptamers (nucleotide fragments engineered to bind regions of targeted molecules). In addition to the improved storage stability, speed of production, and batch-to-batch reproducibility of our DNA aptamers compared to antibodies, we are also able to circumvent intellectual property based on measuring established and/or novel circulating biomarkers using these novel reagents. Beyond the assay technology, the built-in software will enable immediate Health Insurance Portability and Accountability Act (HIPAA) compliant, cloud-based availability of blood test results for access by clinicians during or in follow-up to an e-visit. Therefore, the value to the healthcare system in the management of chronic diseases is multifold, foremost by enhancing patient compliance, comfort, and convenience, while also providing economic value by decreasing the need for costly and time-consuming access to care facilities, staff, and practitioners. Our first application area for TeleLab is heart failure which involves the development of two cartridges: one for assessing kidney health and electrolyte balance (TeleKidney) and one for assessing fluid status (TeleHeart). This application area fits well with the interests of Translational Research Institute for Space Health (TRISH) because the analytes in TeleKidney are sodium, potassium, creatinine, and blood urea nitrogen (BUN) (all deemed high importance or desirable). For TeleHeart, troponin I, Hg/Hct, and glucose are key parts of the multiplexed panel (all deemed high importance or desirable) for measuring fluid status. Our goal for this project is two-fold: develop a second-generation prototype device and cartridge for putting through contract manufacturing in preparation for our Food and Drug Administration (FDA) clinical trials for TeleKidney, and pre-clinically validate a subset (glucose, Hg/Hct, and troponin I) of the assays in the TeleHeart panel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Clinicians strive to decrease hospitalization and mortality rates for heart failure (HF) patients by prescribing trial-proven drug regimens, but face complex decision-making challenges in 1) balancing prescriptions for multiple drug classes and 2) optimizing drug doses and frequencies in treatment strategies which need to be individualized for each patient. All classes of drugs recommended for guidelines-directed medical therapy (GDMT) may cause hypotension or impact renal status by worsening renal function (WRF) and/or electrolyte disorders. Therefore, prescribing any drug or increasing dosage not only adds new potential for precipitating such adverse effects but may also perturb a difficult-to-achieve balance among drugs already in the regimen, thereby jeopardizing existing hemodynamic stability and possibly inducing need for additional compensatory titration in their dosing. Monitoring for these adverse events during a patient's adaptation to any drug change is a core recommendation of GDMT and must also be patient-tailored given their idiosyncratic occurrence. In the home care setting, patients may use devices that measure blood pressure and electronically transmit data to clinicians. However, a critical gap in home monitoring currently exists for renal safety since no available patient-operated device can measure blood levels of renal function and electrolyte parameters and transmit results digitally to care providers. Thus, this information gap may hinder prescribing practices since the real-world concern for renal safety is high for most individuals with HF and comorbidities, in comparison to the relative frequency of adverse renal events observed in studies following HF outpatients. To optimize HF outcomes, the guidelines include recommendations to: 1) add a mineralocorticoid receptor antagonist (MRA); 2) switch to an angiotensin receptor neprilysin inhibitor (ARNI) from angiotensin-converting enzyme inhibitor/angiotensin receptor blocker (ACEI/ARB); and 3) titrate diuretics (e.g., loop and/or thiazide/thiazide-like), and also to achieve trial-proven daily target dose (TD), or an individual's maximum tolerated dose (MTD), for each drug class. All of these drug classes may cause elevations in creatinine (Cr) and/or blood urea nitrogen (BUN) levels, which may either be transient, i.e., associated with reversible fluctuations in renal perfusion, or indicative of renal injury, i.e., impaired glomerular function that may be permanent. Thus, while any rise in Cr is worrisome, monitoring of serial Cr levels is critical to assess risk for WRF and to guide decision-making, with testing frequency proportional to the number and magnitude of prescribed drug changes. These drugs may also cause disorders in the blood levels of potassium (K+) i.e., hyperkalemia or hypokalemia, and/or sodium (Na+), i.e., hyponatremia, which may have immediate and/or fatal dysrhythmic or neurologic consequences. The lack of availability of frequent and convenient blood testing and rapid reporting of these parameters in the usual care for outpatients may be a contributor to renal safety concerns that may limit provider adherence (i.e., "clinician inertia") to GDMT recommendations. Self-care and self-management are longstanding priorities for the HF clinical community and include coordination with providers for proper home administration of drug and dosing regimens. However, renal safety monitoring (RSM) still currently requires engagement with the health system or freestanding laboratory facilities, which may hinder the frequency and/or timeliness of blood testing to guide medication changes and/or dose titration. The InnaMed TeleLab is a novel blood testing device enabling at-home measurement of BUN, Cr, Na+, and K+ to help clinicians remotely assess renal status to safely prescribe additional drugs and up-titrate daily doses to increase GDMT adherence for HF patients.
	The initial proposal deliverables were focused on developing a fully integrated, second-generation TeleLab® system (reader and cartridge) and validating assays for relevant clinical chemistries in spiked and native serum, plasma, and whole blood samples on the benchtop using InnaMed’s electrochemical technology. Though the specific assays and aspects of the system have changed since the initial proposal, the core components of each deliverable are still the same at this time. For the second generation TeleLab system, the key development focused on the integration of the microneedle for sample draw and transfer; automated assay calibration to ensure accuracy and control for electrode-to-electrode variation; and readout via a customized potentiostat that enables a variety of electrochemical measurements for multiplexing. For the microneedle integration, InnaMed started a research collaboration with Seventh Sense Biosystems (7S Bio), a Boston-based company that has developed an FDA-cleared microneedle called TAP® that enables a painless, 200 uL blood draw from the upper arm. This is a key component of InnaMed’s product to enable patient self-testing from the home. In collaboration with 7S Bio and Bresslergroup, InnaMed was able to acquire the microneedle designs and samples to model and design the cartridge for the integration.

Task Progress:	For the assay calibration, InnaMed and Bresslergroup designed a method for calibration using a valve setup with an absorbent pad for pulling solution from the electrode chamber. By using a two-point calibration, InnaMed can ensure accurate results by minimizing potential sensor-to-sensor variation which is especially important in a home setting where the conditions are less controlled. For the potentiostat, InnaMed and Bressler built upon the EmStat Pico, a dual-channel embedded potentiostat/bipotentiostat which can be used as part of a highly compact system for electrochemical systems. This system was a joint development project by PalmSens BV and Analog Devices Inc. and InnaMed and Bressler built additional circuitry upon the platform for multiplexing measurements and additional software algorithms for conducting the two-point calibration and converting the raw data to relevant analyte concentrations. For the assay validation, InnaMed has built upon the cartridge design and developed customized flow cells for measurement. The flow cells are an important step towards a final product because they help to show what a raw data trace would look like in the cartridge with the sample flowing. This is a required step to understand the on-chip reaction kinetics and analytical performance to help improve the software algorithms. This testing has been done for the renal function and electrolyte assays and will be expanded to the DNA-based assays. Further testing will then be done in collaboration with the Hospital at the University of Pennsylvania in a clinical study using whole blood samples collected by the microneedle-integrated cartridges for validation prior to FDA clinical trials. Overall, InnaMed has made substantial progress on this project toward an FDA-cleared TeleLab system and cartridge initially for heart failure.
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